

ELECTRIC CHARGES AND FIELDS

Important formulas and concepts directly from the NCERT-

❖ **Electrostatics** – branch of physics which studies about charge and its properties like electric field when it is at rest.

❖ **Charge** – it is the property of a particle due to which it produces or experiences electric and magnetic effects.

SI unit – coulomb

Dimensions – [AT]

❖ Properties of charge-

➤ **Charge is added algebraically.**

➤ **Charge on a body is quantised.**

$$Q = ne$$

➤ **Net charge of an isolated system is conserved.**

➤ 1) charge at rest produce electric fields(E).

2) charge in uniform motion produce both electric and magnetic fields(B).

3) accelerated charge produce E,B and EM waves.

❖ Coulomb's Law-

➤ Electrostatic force b/w two charges,

$$F = kq_1q_2/r^2$$

$$(k = 1/4\pi\epsilon_0)$$

k – depend on medium

$$\text{value} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

ϵ_0 = permittivity of free space

$$\text{value} = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$\text{dimension} = [\text{M}^{-1}\text{L}^{-3}\text{T}^4\text{A}^2]$$

➤ Properties of electrostatic force –

○ **Applicable for point charges.**

○ **Conservative in nature.**

○ **Follow inverse square law.**

❖ Principle of superposition-

➤ **Force b/w any two charges is independent of presence of other charges.**

➤ **Net force on any charge is vector sum of forces acting on it due to all other charges.**

Note- net force on charge present at geometrical centre due to identical charges present at corners is ‘zero’.

❖ **Effect of medium on coulomb’s force –**

➤ Force in medium,

$$F_m = q_1 q_2 / 4\pi\epsilon_m r^2$$

ϵ_m is the permittivity of medium.

➤ **Relative permittivity,**

$$\epsilon_r = k = \text{dielectric constant}$$

$$k = \epsilon_m / \epsilon_0$$

$$\epsilon_m = \epsilon_0 \times k$$

➤ Force in medium,

$$F_m = F_o / k$$

(F_o = force in air

F_m = force in medium)

Here, k = dielectric constant

❖ **Electric field –**

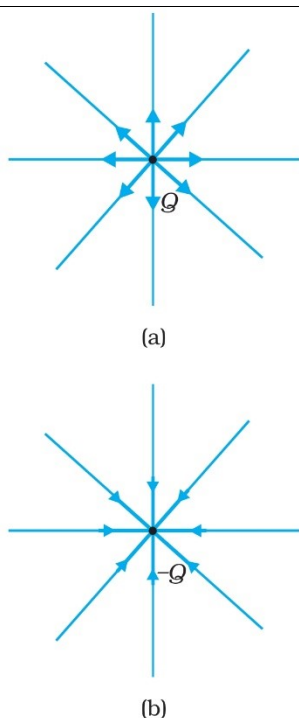


FIGURE 1.11 Electric field (a) due to a charge Q , (b) due to a charge $-Q$.

➤ Electric field due to point charge,

$$E = kQ / r^2$$

➤ Electric field due to

Positive charge – away from charge

Negative charge – towards the charge

➤ Electric field at centre due to uniformly charged ring – **zero**

➤ Electric field on the axis of a ring,

$$E = kQx / (R^2 + x^2)^{3/2}$$

○ If $x \ll R$, $E = kQx / R^3$

○ If $x \gg R$, $E = kQ / x^2$

➤ Electric field due to uniformly charged arc,

$$E = 2k\lambda \sin(\Theta/2) / R$$

Electric field due to semi-circular arc,

$$E = 2k\lambda / R$$

➤ Electric field due to uniformly charged **Infinite wire** = $2k\lambda / R$

Force b/w two infinitely charged wire,

$$F/l = 2k\lambda_1\lambda_2 / R$$

➤ Electric field due to uniformly charged cylinder,

- **Hollow cylinder** - **inside, $E = 0$**
 outside, $E = 2k\lambda / R$

- **Solid cylinder** - **inside, $E \propto R$**
 Outside, $E = 2k\lambda / R$

➤ electric field due to infinite uniform charged sheet –

- **Non-conducting sheet,** $E = \sigma / 2\epsilon_0 = Q / 2A\epsilon_0$
- **Conducting sheet,** $E = \sigma / \epsilon_0 = Q / 2A\epsilon_0$

➤ Electric field due to sphere-

1) Conducting solid / hollow sphere or non-conducting hollow sphere,

$$\begin{array}{ll} \text{Inside,} & E = 0 \\ \text{At surface,} & E = kQ / R^2 \\ \text{Outside,} & E = kQ / r^2 \quad (r > R) \end{array}$$

2) Uniformly charged sphere or non-conducting solid sphere,

$$\text{Inside,} \quad E = kQr / R^3$$

$$\begin{array}{ll} \text{At surface,} & E = kQ / R^2 \\ \text{Outside,} & E = kQ / r^2 \quad (r > R) \end{array}$$

- A uniformly charged sphere and a cavity is made inside it, Then field inside the cavity is 'constant and uniform'.

❖ Electric Dipole-

- Two charges of same magnitude and opposite nature separated by some distance forms an electric dipole.

- $\mathbf{P} = q \times l$

- Direction of dipole – (-q to +q)

- SI unit – C.m

- Electric field due to dipole,

- 1) At axial position, $E = 2kP / r^3$

- 2) At equatorial position, $E = kP / r^3$

(electric field at equatorial position is always opposite to the direction of dipole)

- 3) At any general point, $E = kP (1+3\cos^2\Theta)^{1/2} / r^3$

- Force on a charge due to dipole,

$$F \propto 1/r^3$$

- Net force on dipole in uniform electric field is 'zero'.

❖ Electric flux-

- no. of electric field lines passing through any surface.

$$\Phi = \mathbf{E} \cdot \mathbf{A}$$

- properties of electric field lines-

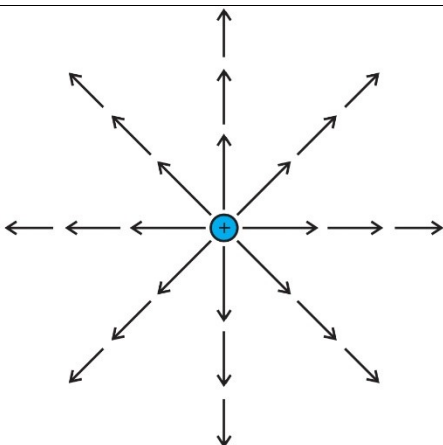


FIGURE 1.15 Field of a point charge.

- in a charge free region , field lines can be taken to continuous curves without any breaks.
- hypothetical lines to represent electric field.
- For positive charge, start from $+q$ and ends at infinity.
For negative charge, start from infinity and ends at $-q$.
- “tangent drawn on any point of electric field lines will give the direction of electric field at that point”.
- Electric field is more where field lines density is more.
- Electric field is conservative in nature , never forms a close loop.
- Electric field lines never intersect each other, because there is two direction of electric field at a point which is not possible.
- They are always perpendicular to the surface of conductor.

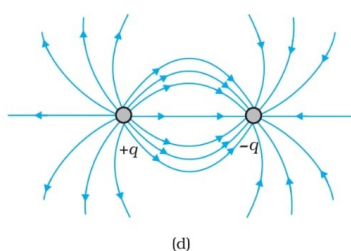
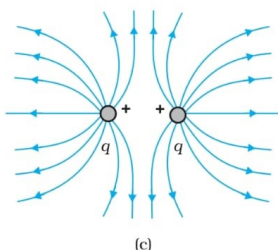
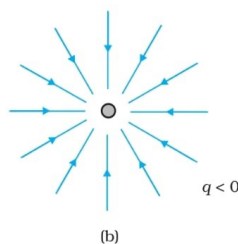
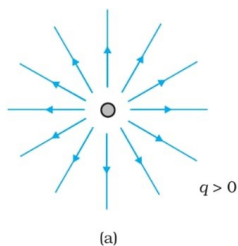


FIGURE 1.17 Field lines due to some simple charge configurations.

➤ Acc. To this law, **“The net flux passing through any closed surface is $1/\epsilon_0$ times the total charge enclosed inside the gaussian surface”**.

➤ Properties of Gauss Law-

- ### ❖ Motion of charged particle inside uniform electric field –

- **Particle's acceleration,** $\mathbf{a} = \mathbf{qE} / m$
Velocity, $\mathbf{v} = (\mathbf{qE} / m) \mathbf{t}$
Displacement, $\mathbf{x} = \frac{1}{2} (\mathbf{qE} / m) \mathbf{t}^2$

- **Highest specific charge – maximum deviation**

SUMMARY

1. Electric and magnetic forces determine the properties of atoms, molecules and bulk matter.
2. From simple experiments on frictional electricity, one can infer that there are two types of charges in nature; and that like charges repel and unlike charges attract. By convention, the charge on a glass rod rubbed with silk is positive; that on a plastic rod rubbed with fur is then negative.
3. Conductors allow movement of electric charge through them, insulators do not. In metals, the mobile charges are electrons; in electrolytes both positive and negative ions are mobile.
4. Electric charge has three basic properties: quantisation, additivity and conservation.

Quantisation of electric charge means that total charge (q) of a body is always an integral multiple of a basic quantum of charge (e) i.e., $q = n e$, where $n = 0, \pm 1, \pm 2, \pm 3, \dots$. Proton and electron have charges $+e, -e$, respectively. For macroscopic charges for which n is a very large number, quantisation of charge can be ignored.

Additivity of electric charges means that the total charge of a system is the algebraic sum (i.e., the sum taking into account proper signs) of all individual charges in the system.

Conservation of electric charges means that the total charge of an isolated system remains unchanged with time. This means that when

bodies are charged through friction, there is a transfer of electric charge from one body to another, but no creation or destruction of charge.

5. *Coulomb's Law*: The mutual electrostatic force between two point charges q_1 and q_2 is proportional to the product $q_1 q_2$ and inversely proportional to the square of the distance r_{21} separating them. Mathematically,

$$\mathbf{F}_{21} = \text{force on } q_2 \text{ due to } q_1 = \frac{k (q_1 q_2)}{r_{21}^2} \hat{\mathbf{r}}_{21}$$

where $\hat{\mathbf{r}}_{21}$ is a unit vector in the direction from q_1 to q_2 and $k = \frac{1}{4\pi\epsilon_0}$ is the constant of proportionality.

In SI units, the unit of charge is coulomb. The experimental value of the constant ϵ_0 is

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

The approximate value of k is

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

6. The ratio of electric force and gravitational force between a proton and an electron is

$$\frac{k e^2}{G m_e m_p} \cong 2.4 \times 10^{39}$$

7. *Superposition Principle*: The principle is based on the property that the forces with which two charges attract or repel each other are not affected by the presence of a third (or more) additional charge(s). For an assembly of charges $q_1, q_2, q_3, \dots$, the force on any charge, say q_1 , is the vector sum of the force on q_1 due to q_2 , the force on q_1 due to q_3 , and so on. For each pair, the force is given by the Coulomb's law for two charges stated earlier.
8. The electric field $\mathbf{E}$ at a point due to a charge configuration is the force on a small positive test charge q placed at the point divided by the magnitude of the charge. Electric field due to a point charge q has a magnitude $|q|/4\pi\epsilon_0 r^2$; it is radially outwards from q , if q is positive, and radially inwards if q is negative. Like Coulomb force, electric field also satisfies superposition principle.
9. An electric field line is a curve drawn in such a way that the tangent at each point on the curve gives the direction of electric field at that point. The relative closeness of field lines indicates the relative strength of electric field at different points; they crowd near each other in regions of strong electric field and are far apart where the electric field is weak. In regions of constant electric field, the field lines are uniformly spaced parallel straight lines.
10. Some of the important properties of field lines are: (i) Field lines are continuous curves without any breaks. (ii) Two field lines cannot cross each other. (iii) Electrostatic field lines start at positive charges and end at negative charges—they cannot form closed loops.
11. An electric dipole is a pair of equal and opposite charges q and $-q$ separated by some distance $2a$. Its dipole moment vector $\mathbf{p}$ has magnitude $2qa$ and is in the direction of the dipole axis from $-q$ to q .

12. Field of an electric dipole in its equatorial plane (i.e., the plane perpendicular to its axis and passing through its centre) at a distance r from the centre:

$$\mathbf{E} = \frac{-\mathbf{p}}{4 \pi \epsilon_0} \frac{1}{(a^2 + r^2)^{3/2}}$$

$$\cong \frac{-\mathbf{p}}{4 \pi \epsilon_0 r^3}, \quad \text{for } r \gg a$$

Dipole electric field on the axis at a distance r from the centre:

$$\mathbf{E} = \frac{2 \mathbf{p} r}{4 \pi \epsilon_0 (r^2 - a^2)^2}$$

$$\cong \frac{2 \mathbf{p}}{4 \pi \epsilon_0 r^3} \quad \text{for } r \gg a$$

The $1/r^3$ dependence of dipole electric fields should be noted in contrast to the $1/r^2$ dependence of electric field due to a point charge.

13. In a uniform electric field $\mathbf{E}$, a dipole experiences a torque $\boldsymbol{\tau}$ given by

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$$

but experiences no net force.

14. The flux $\Delta\phi$ of electric field $\mathbf{E}$ through a small area element $\Delta\mathbf{S}$ is given by

$$\Delta\phi = \mathbf{E} \cdot \Delta\mathbf{S}$$

The vector area element $\Delta\mathbf{S}$ is

$$\Delta\mathbf{S} = \Delta S \hat{\mathbf{n}}$$

where ΔS is the magnitude of the area element and $\hat{\mathbf{n}}$ is normal to the area element, which can be considered planar for sufficiently small ΔS .

For an area element of a closed surface, $\hat{\mathbf{n}}$ is taken to be the direction of *outward* normal, by convention.

15. *Gauss's law*: The flux of electric field through any closed surface S is $1/\epsilon_0$ times the total charge enclosed by S . The law is especially useful in determining electric field $\mathbf{E}$, when the source distribution has simple symmetry:

(i) *Thin infinitely long straight wire of uniform linear charge density λ*

$$\mathbf{E} = \frac{\lambda}{2 \pi \epsilon_0 r} \hat{\mathbf{n}}$$

where r is the perpendicular distance of the point from the wire and $\hat{\mathbf{n}}$ is the radial unit vector in the plane normal to the wire passing through the point.

(ii) *Infinite thin plane sheet of uniform surface charge density σ*

$$\mathbf{E} = \frac{\sigma}{2 \epsilon_0} \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the plane, outward on either side.

(iii) *Thin spherical shell of uniform surface charge density σ*

$$\mathbf{E} = \frac{q}{4 \pi \epsilon_0 r^2} \hat{\mathbf{r}} \quad (r \geq R)$$

$$\mathbf{E} = 0 \quad (r < R)$$

where r is the distance of the point from the centre of the shell and R the radius of the shell. q is the total charge of the shell: $q = 4\pi R^2 \sigma$.

The electric field outside the shell is as though the total charge is concentrated at the centre. The same result is true for a solid sphere of uniform volume charge density. The field is zero at all points inside the shell.

POINTS TO PONDER

1. You might wonder why the protons, all carrying positive charges, are compactly residing inside the nucleus. Why do they not fly away? You will learn that there is a third kind of a fundamental force, called the strong force which holds them together. The range of distance where this force is effective is, however, very small $\sim 10^{-14}$ m. This is precisely the size of the nucleus. Also the electrons are not allowed to sit on top of the protons, i.e. inside the nucleus, due to the laws of quantum mechanics. This gives the atoms their structure as they exist in nature.
2. Coulomb force and gravitational force follow the same inverse-square law. But gravitational force has only one sign (always attractive), while

Coulomb force can be of both signs (attractive and repulsive), allowing possibility of cancellation of electric forces. This is how gravity, despite being a much weaker force, can be a dominating and more pervasive force in nature.

3. The constant of proportionality k in Coulomb's law is a matter of choice if the unit of charge is to be defined using Coulomb's law. In SI units, however, what is defined is the unit of current (A) via its magnetic effect (Ampere's law) and the unit of charge (coulomb) is simply defined by ($1\text{C} = 1\text{ A s}$). In this case, the value of k is no longer arbitrary; it is approximately $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.
4. The rather large value of k , i.e., the large size of the unit of charge (1C) from the point of view of electric effects arises because (as mentioned in point 3 already) the unit of charge is defined in terms of magnetic forces (forces on current-carrying wires) which are generally much weaker than the electric forces. Thus while 1 ampere is a unit of reasonable size for magnetic effects, $1\text{ C} = 1\text{ A s}$, is too big a unit for electric effects.
5. The additive property of charge is not an 'obvious' property. It is related to the fact that electric charge has no direction associated with it; charge is a scalar.
6. Charge is not only a scalar (or invariant) under rotation; it is also invariant for frames of reference in relative motion. This is not always true for every scalar. For example, kinetic energy is a scalar under rotation, but is not invariant for frames of reference in relative motion.
7. Conservation of total charge of an isolated system is a property independent of the scalar nature of charge noted in point 6. Conservation refers to invariance in time in a given frame of reference. A quantity may be scalar but not conserved (like kinetic energy in an inelastic collision). On the other hand, one can have conserved vector quantity (e.g., angular momentum of an isolated system).
8. Quantisation of electric charge is a basic (unexplained) law of nature; interestingly, there is no analogous law on quantisation of mass.
9. Superposition principle should not be regarded as 'obvious', or equated with the law of addition of vectors. It says two things: force on one charge due to another charge is unaffected by the presence of other charges, and there are no additional three-body, four-body, etc., forces which arise only when there are more than two charges.
10. The electric field due to a discrete charge configuration is not defined at the locations of the discrete charges. For continuous volume charge distribution, it is defined at any point in the distribution. For a surface charge distribution, electric field is discontinuous across the surface.
11. The electric field due to a charge configuration with total charge zero is not zero; but for distances large compared to the size of the configuration, its field falls off faster than $1/r^2$, typical of field due to a single charge. An electric dipole is the simplest example of this fact.

